Lucas Coefficient Families and Exact Arithmetic for Every Integer Coefficient

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Abstract

We study weighted Lucas expressions for positive integer coefficients. Every positive difference $L_i - 5F_{2r}L_j$, for $r \geq 1$, has a unique representation; we determine its admissible indices, bounded-index count, and asymptotics. In contrast, for every $q = L_{2r}$, both $qL_i - L_j$ and $qL_i + L_j$ have infinitely many multiply represented positive values. We prove broader quotient and half-Lucas criteria, including an infinite family of triple representations. A transfer lemma relates sum collisions to reversed-difference collisions. An algebraic-norm bound gives exact representation and rank algorithms for every integer coefficient, and excludes all positive simultaneous-shift collision families for coefficients at least two. This exclusion and an effective algebraic-norm estimate also yield a finite upper bound on every positive collision, deciding complete uniqueness for each fixed integer coefficient at least two.

Let $F_0 = 0, F_1 = 1$ and $L_0 = 2, L_1 = 1$, with $u_{n+2} = u_{n+1} + u_n$. Use $F_{-n} = (-1)^{n+1}F_n$ and $L_{-n} = (-1)^n L_n$. Put

$$S_q = \{L_i - qL_j > 0 : i, j \geq 1\}, \quad T_q = \{qL_i - L_j > 0 : i, j \geq 1\}, \quad P_q = \{qL_i + L_j : i, j \geq 1\}.$$

The recurrence gives

$$L_{m+4} - L_m = 5F_{m+2}, \qquad F_a L_b = F_{a+b} + (-1)^b F_{a-b}. \tag{1}$$

The collision-free coefficients begin $5, 15, 40, 105, 275, \ldots$; the persistent-collision coefficients begin $3, 7, 18, 47, 123, \ldots$.

An infinite collision-free family

Theorem 1. *Fix $r \geq 1$ and set $q = 5F_{2r}$. Then*

$$L_i - qL_j > 0 \quad \Longleftrightarrow \quad (i, j) = (2r + 2, 1) \quad \text{or} \quad i \geq 2r + 3,\ 1 \leq j \leq i - 2r - 2.$$

Every positive integer in S_q has precisely one representation.

Proof. The recurrence yields

$$q = 5F_{2r} = L_{2r+2} - L_{2r-2}, \quad q - L_{2r+1} = L_{2r-1} > 0, \quad L_{2r+2} - q = L_{2r-2} > 0. \tag{2}$$

Thus $j = 1$ first becomes admissible at $i = 2r + 2$. For $j \geq 2$ put $A_j = L_{j+2r+2} - qL_j$ and $B_j = qL_j - L_{j+2r+1}$. Both obey the Lucas recurrence in j. Checking two initial values in r and using the recurrence in steps of two gives

$$A_2 = L_{2r+4} - 3q = L_{2r-4} > 0, \quad A_3 = A_2 + A_1 > 0, \quad A_1 = L_{2r+3} - q > 0,$$

with $L_{-2} = 3$ when $r = 1$. For B_j, the first two terms of the sequences $B_2(r)$ and $B_3(r)$ are

$$(B_2(1), B_2(2)) = (4, 16), \qquad (B_3(1), B_3(2)) = (2, 13).$$

Each satisfies $b_{r+2} = 3b_{r+1} - b_r$ as a function of r. An increasing positive initial pair remains increasing and positive. Therefore $B_2, B_3 > 0$ for all $r \geq 1$; the recurrence in j then gives $A_j, B_j > 0$ for all $j \geq 2$. Hence $L_{j+2r+1} < qL_j < L_{j+2r+2}$ for $j \geq 2$, establishing the stated index region.

Let $\mathcal{B}_i$ be the positive values with first index i. They are distinct within each block, and $\max \mathcal{B}_i = L_i - q$ for $i \geq 2r + 2$. Using (1) with $b = i - 2r + 2$, the next block of the same residue class satisfies

$$\begin{aligned} \min \mathcal{B}_{i+4} - \max \mathcal{B}_i &= L_{i+4} - L_i - qL_{i-2r+2} + q \\ &= q + 5F_{i-4r+2} > 0. \end{aligned} \tag{3}$$

For a nonnegative Fibonacci index the inequality is immediate. For a negative index, $i \geq 2r + 2$ implies $0 < |i - 4r + 2| \leq 2r - 4 < 2r$, so $|F_{i-4r+2}| < F_{2r}$. Moreover $A_{j+4} > A_j$ follows from the strictly positive Lucas recurrence; thus blocks separated by any positive multiple of four are strictly ordered.

Finally, $(L_1, L_2, L_3, L_4) \bmod 5 = (1, 3, 4, 2)$ is the repeating residue cycle. Since $5 \mid q$, two equal differences must have first indices congruent modulo four. They cannot belong to different such blocks, by (3), and the same block has distinct values. □

Corollary 2. *For $N \geq 2r + 2$ the number of distinct positive differences with first index at most N equals*

$$1 + \frac{(N - 2r - 2)(N - 2r - 1)}{2}.$$

Writing $\varphi = (1 + \sqrt{5})/2$ and $A_r(x) = \#(S_{5F_{2r}} \cap [1, x])$,

$$A_r(x) = \tfrac{1}{2}(\log_\varphi x)^2 + O_r(\log x), \qquad \log_\varphi b_{r,n} = \sqrt{2n} + O_r(1),$$

where $b_{r,n}$ is the n-th positive value. Its exact value can be recovered by integer binary search.

Proof. Summing block sizes $1, 1, 2, 3, \ldots$ gives the formula. Let $m(x) = \max\{m \geq 1 : L_m \leq x\} = \log_\varphi x + O(1)$. All positive differences with $i \leq m(x)$ are at most x. For $j \geq 2$, $A_2 \geq L_0 = 2$ and $A_3 \geq L_1 = 1$, so recurrence gives $A_j \geq L_{j-2}$. Therefore, for all sufficiently large i, a value in $\mathcal{B}_i$ is at least L_{i-2r-4}; if it is at most x then $i \leq m(x) + 2r + 4$. Applying the triangular count to these two index bounds proves the asymptotic. Inverting it proves the nth-term estimate. An exact count tests $L_j \geq \lceil (L_i - x)/q \rceil$ within finitely many admissible blocks; binary search inverts that monotone count. □

A family with infinitely many collisions

Theorem 3. *For every $r \geq 1$, the coefficient $q = L_{2r}$ has infinitely many repeated positive values in both T_q and P_q. For every $n \geq 1$,*

$$L_{n+4r} = qL_{n+2r} - L_n = qL_{n+6r} - L_{n+8r} > 0, \tag{4}$$

$$qL_{n+2r} + L_{n+8r} = qL_{n+6r} + L_n. \tag{5}$$

The two index pairs in each equality are distinct.

Proof. The product identity $L_aL_b = L_{a+b} + (-1)^a L_{b-a}$ with $a = 2r$ gives

$$qL_{n+2r} = L_{n+4r} + L_n, \quad qL_{n+6r} = L_{n+8r} + L_{n+4r}.$$

These yield (4) and (5). Both resulting sequences of positive values increase strictly with n, providing infinitely many collisions. □

Proposition 4 (Collision transfer)**.** *Suppose S_q has unique positive representations and q divides no Lucas number L_j with $j \geq 1$. Then every sum collision $qL_a + L_b = qL_c + L_d$ with $a < c$ induces a positive collision $qL_a - L_d = qL_c - L_b > 0$ in T_q. These hypotheses hold for every $q = 5F_{2r}$.*

Proof. Set $t = qL_a - L_d = qL_c - L_b$. If $t = 0$, then $q \mid L_d$, a contradiction. If $t < 0$, the equality $L_d - qL_a = L_b - qL_c > 0$ gives two representations in S_q, also a contradiction. Thus $t > 0$. The final assertion follows from Theorem 1 and the four nonzero Lucas residues modulo five. □

Divisibility families of infinite collisions

Theorem 5 (Lucas–Fibonacci quotient coefficients)**.** *Let $m > h \geq 1$ have the same parity, and suppose*

$$q = \begin{cases} L_m/L_h \in \mathbb{Z}, & m, h \text{ odd}, \\ F_m/F_h \in \mathbb{Z}, & m, h \text{ even}. \end{cases}$$

Then $q \geq 2$, and for every $n \geq 1$,

$$qL_{n+m-h} - L_n = qL_{n+m+h} - L_{n+2m} > 0. \tag{6}$$

In particular, T_q and P_q each contain infinitely many positive values having two distinct representations. The coefficient $q = 2$ also has infinitely many collisions in both sets.

Proof. For m odd, the Lucas product formula gives

$$L_{n+2m} - L_n = L_mL_{n+m}, \qquad L_{n+m+h} - L_{n+m-h} = L_hL_{n+m}.$$

For m, h even, the Lucas–Fibonacci identity instead gives

$$L_{n+2m} - L_n = 5F_mF_{n+m}, \quad L_{n+m+h} - L_{n+m-h} = 5F_hF_{n+m}.$$

In either case dividing the two identities gives

$$L_{n+2m} - L_n = q(L_{n+m+h} - L_{n+m-h}),$$

which is (6). The right-hand side of the first expression $qL_{n+m-h} - L_n$ is strictly positive and increases with n because $q \geq 2$ and $m - h \geq 1$. The equivalent sum collision is

$$qL_{n+m-h} + L_{n+2m} = qL_{n+m+h} + L_n.$$

For $q = 2$, the recurrence gives the further family

$$2L_{n+2} - L_n = L_{n+3} = 2L_{n+3} - L_{n+3},$$

again with strictly increasing positive values. The corresponding sum identity follows by swapping the two subtracted indices. □

Remark 6. *The theorem covers* every *Lucas coefficient $q = L_k$ ($k \geq 1$): for odd k choose $(m, h) = (k, 1)$ when $k > 1$, and for even $k = 2r$ choose $(m, h) = (4r, 2r)$ using $F_{4r}/F_{2r} = L_{2r}$. For $q = L_1 = 1$, $L_{n+2} - L_n = L_{n+3} - L_{n+2} > 0$ already provides infinitely many collisions. It also covers infinitely many other integer coefficients, for example $8, 19, 21, 48, 55$.*

Theorem 7 (Half-Lucas collision coefficients)**.** *For every integer $r \geq 0$, let $M = 6r + 3$ and*

$$q = \frac{L_M}{2} \in \mathbb{Z}.$$

Then T_q and P_q each have infinitely many repeated positive values. More precisely, for every $x \geq M$,

$$qL_x - L_{x-M+1} = qL_{x+3} - L_{x+M+1} > 0, \tag{7}$$

and

$$qL_x + L_{x+M+1} = qL_{x+3} + L_{x-M+1}.$$

The coefficients begin $2, 38, 682, 12238, 219602, \ldots$.

Proof. The Lucas residues modulo 2 have period 3, so L_M is even when $3 \mid M$. As M is odd, the product identity gives

$$L_{x+M+1} - L_{x-M+1} = L_M L_{x+1}.$$

The recurrence also gives $L_{x+3} - L_x = 2L_{x+1}$. Substituting $2q = L_M$ proves (7) and the sum identity. As $x \geq M \geq 3$, $qL_x - L_{x-M+1} > 0$; these values strictly increase with x. □

Corollary 8 (Triple collisions for $q = 2$)**.** *For every $n \geq 3$,*

$$\boxed{L_{n+1} = 2L_n - L_{n-2} = 2L_{n+1} - L_{n+1} = 2L_{n+3} - L_{n+4}.}$$

Thus T_2 has infinitely many values with at least three distinct representations.

Proof. The first equality follows from $L_{n+1} = 2L_n - L_{n-2}$, the second is immediate, and the third is Theorem 7 with $r = 0$. □

An exact algorithm for every integer coefficient

The coefficient families above admit efficient special-purpose rankings. An integer-bound argument gives complete, exact arithmetic for every coefficient, with no conjectural collision classification needed.

Theorem 9 (Uniform finite search bounds)**.** *Let $q \geq 1$ and $X \geq 1$ be integers, and put*

$$B = (q + 1)(X + q + 1) + 1.$$

For any $0 < L_i - qL_j \leq X$ with $i, j \geq 1$, necessarily

$$L_j \leq B, \qquad L_i \leq X + qB. \tag{8}$$

For $q \geq 2$ and $0 < qL_i - L_j \leq X$, necessarily

$$L_i \leq M := \max\{B, \lfloor X/(q-1) \rfloor\}, \qquad L_j \leq qM. \tag{9}$$

When $q = 1$, the expressions $qL_i - L_j$ and $L_i - qL_j$ coincide and obey (8). For $q \geq 1$ and $0 < qL_i + L_j \leq X$,

$$L_i \leq \lfloor (X-1)/q \rfloor, \qquad L_j \leq X - q. \tag{10}$$

Consequently, for every *$q \in \mathbb{Z}$, the set of distinct positive values at most X in each of S_q, T_q, P_q is effectively computable by finite integer arithmetic. For every nonzero q, all representations of each positive value can also be enumerated, yielding its exact multiplicity. For $q = 0$, the positive sets S_0 and P_0 are the positive-index Lucas numbers, each with infinitely many representations; T_0 is empty.*

Proof. Write $\alpha = \varphi$ and $\beta = (1 - \sqrt{5})/2 = -1/\varphi$. For $d \geq 1$ and $q \geq 1$, the algebraic integer $\alpha^d - q$ has nonzero integral norm

$$(\alpha^d - q)(\beta^d - q) \in \mathbb{Z} \setminus \{0\}.$$

It cannot vanish because $\alpha^d = F_d\alpha + F_{d-1}$ is irrational. Since $|\beta^d - q| \leq q + 1$, this gives

$$|\alpha^d - q| \geq \frac{1}{q+1}. \tag{11}$$

A positive difference $L_i - qL_j$ has $i > j$; write $d = i - j$. By Binet's formula,

$$L_i - qL_j = \alpha^j(\alpha^d - q) + \beta^i - q\beta^j.$$

For $0 < L_i - qL_j \leq X$, it follows that

$$\frac{\alpha^j}{q+1} \leq X + |\beta^i - q\beta^j| \leq X + q + 1.$$

Thus $\alpha^j \leq (q+1)(X+q+1)$, and $L_j = \alpha^j + \beta^j \leq B$. The equality $L_i = x + qL_j$ now proves (8).

For $0 < qL_i - L_j \leq X$, when $j > i$, the same argument with $d = j - i$ shows $L_i \leq B$, hence $L_j = qL_i - x \leq qB$. When $j \leq i$ and $q \geq 2$, $x = qL_i - L_j \geq (q-1)L_i$; this proves (9). The sum bounds (10) follow from $L_i, L_j \geq 1$.

For negative coefficients, write $m = -q > 0$. Then $S_{-m} = P_m$ with the indices interchanged, $P_{-m} = S_m$ with the indices interchanged, and T_{-m} is empty. The assertions for $q = 0$ are immediate. Thus the finite index bounds cover all integers. □

Corollary 10 (Universal exact counting and unranking)**.** *For any integer q, expression type S, T or P, and integer $X \geq 0$, there is a terminating algorithm for the number of* distinct *positive values at most X, exact membership, and all positive representations (or detection of infinite multiplicity when $q = 0$). Whenever the positive set is nonempty, its n-th distinct value is computable exactly for every $n \geq 1$.*

Proof. Enumerate the finitely many indices specified by Theorem 9, retain the positive values at most X, and deduplicate integer values. Each nonempty positive set is unbounded: for S_q with $q > 0$ fix $j = 1$ and let $i \to \infty$, for T_q with $q > 0$ fix $j = 1$ and let $i \to \infty$, and for P_q with $q > 0$ fix $i = 1$ and let $j \to \infty$. The negative and zero cases reduce to these or the Lucas numbers themselves. Therefore doubling the threshold until n values are found and then binary-searching the exact count always terminates. The only empty case is T_q for $q \leq 0$. □

Proposition 11 (No simultaneous-shift positive collisions)**.** *For every integer $q \geq 2$, there are no indices $i < k$, $j < \ell$ such that*

$$L_{n+i} - qL_{n+j} = L_{n+k} - qL_{n+\ell} > 0 \quad \textit{for every } n \geq 0.$$

Proof. An equality for all shifts implies, by Binet's formula, both its α- and β-coefficient equalities. Put $h = k - i \geq 1$, $e = \ell - j \geq 1$, and $s = i - j$. Positivity for large n forces $\alpha^s > q$, so $s \geq 2$, and

$$\alpha^s(\alpha^h - 1) = q(\alpha^e - 1)$$

forces $e > h$, hence $e \geq 2$. Conjugation gives

$$q|\beta^e - 1| = |\beta|^s|\beta^h - 1| \leq \alpha^{-2}(1 + \alpha^{-1}) = \alpha^{-1}.$$

But $q|\beta^e - 1| \geq 2(1 - \alpha^{-2}) = 2/\alpha > \alpha^{-1}$, a contradiction. □

Finite global collisions for every coefficient $q \geq 2$

A finite threshold is sufficient to compute any term. The stronger statement below provides a bound independent of the value.

Theorem 12 (Uniform complete collision certificate)**.** *Fix an integer $q \geq 2$. Let $t = t(q)$ be the least positive index such that*

$$L_t > 200(q + 1)^4. \tag{12}$$

If two distinct positive representations satisfy

$$L_i - qL_j = L_k - qL_\ell > 0, \qquad i < k,$$

then

$$\boxed{k \leq 6t.} \tag{13}$$

Consequently every S_q with $q \geq 2$ has only finitely many multiply represented positive values. Its complete collision list and global uniqueness can be determined by checking the finitely many pairs $1 \leq j < i \leq 6t$.

Proof. Write $\alpha = \varphi$, $\beta = -\alpha^{-1}$. The equality forces $j < \ell$. Moreover, positivity gives $k > \ell$. Put $d = k - \ell$, $h = k - i$ and $g = k - j$. We prove $k \leq 6t$ by contradiction. The bounds

$$\frac{1}{2}\alpha^n \leq L_n \leq 2\alpha^n \quad (n \geq 0), \qquad L_k - L_i \geq L_{k-2}$$

follow from Binet's formula and monotonicity. Since

$$L_{k-2} \leq L_k - L_i = q(L_\ell - L_j) < qL_\ell,$$

we have $\alpha^d < 4q\alpha^2 < \alpha^t$; thus $1 \leq d < t$. Suppose $k > 6t$. Then $\ell = k - d > 5t$.

Write

$$x = L_k - qL_\ell = \alpha^\ell(\alpha^d - q) + \beta^k - q\beta^\ell.$$

The norm argument in (11) gives $|\alpha^d - q| \geq 1/(q + 1)$. Since $|\beta^k - q\beta^\ell| \leq q + 1$ and $\alpha^\ell > 2(q + 1)^2$, the positivity of x forces $\alpha^d - q > 0$ and

$$x \geq \frac{\alpha^\ell}{2(q + 1)}.$$

But $x < L_i \leq 2\alpha^i$, so $\alpha^{\ell - i} < 4(q + 1) < \alpha^t$, which gives $1 \leq h < 2t$.

Next, the original equality can be rearranged as

$$qL_j = L_i - L_k + qL_\ell = \alpha^k A + \beta^k A',$$

where

$$A = \alpha^{-h} - 1 + q\alpha^{-d}, \qquad A' = \beta^{-h} - 1 + q\beta^{-d}.$$

Both are conjugate algebraic integers, and

$$|A'| \le \alpha^h + 1 + q\alpha^d < 3\alpha^{2t} < \alpha^{3t}.$$

If $A = 0$, then $qL_j = |\beta^k A'| < 1$, an impossibility. Otherwise AA' is a nonzero integer, so $|A| \ge \alpha^{-3t}$. As $k > 6t$ and $qL_j > 0$, this forces $A > 0$ and

$$qL_j > \tfrac{1}{2}\alpha^{k-3t}.$$

Using $qL_j \le 2q\alpha^j$ gives $\alpha^{k-j} < 4q\alpha^{3t} < \alpha^{4t}$, hence $g < 4t$.

Finally, express the collision as

$$0 = L_k - L_i - qL_\ell + qL_j = \alpha^k E + \beta^k E',$$

with the conjugate algebraic integers

$$E = 1 - \alpha^{-h} - q\alpha^{-d} + q\alpha^{-g}, \quad E' = 1 - \beta^{-h} - q\beta^{-d} + q\beta^{-g}.$$

The bounds on d, h, g imply

$$|E'| \le 1 + \alpha^{2t} + q\alpha^t + q\alpha^{4t} < \alpha^{6t}.$$

If $E \ne 0$, the integral norm $EE' \ne 0$ gives $|E| \ge 1/|E'|$. The displayed equality now implies $\alpha^{2k} \le |E'|^2 < \alpha^{12t}$, contrary to $k > 6t$.

If $E = 0$, the equality also forces $E' = 0$. It then holds after every simultaneous translation of all four indices. Because $\alpha^d - q > 0$, the translated differences are positive for all sufficiently large translations, contradicting Proposition 11. Thus neither case is possible, and $k \le 6t$. □

Corollary 13 (Complete coefficient-by-coefficient uniqueness decisions and growth)**.** *For any input integer $q \ge 2$, a finite, exact integer algorithm decides whether every positive value $L_i - qL_j$ is uniquely represented. If not, it returns every multiply represented value together with all its index pairs. Furthermore, for each fixed $q \ge 2$,*

$$\#\{x \le X : x \in S_q\} = \tfrac{1}{2}(\log_\varphi X)^2 + O_q(\log X).$$

Proof. All first indices of a collision are at most $6t$ by Theorem 12; finite enumeration therefore gives a complete answer. For the asymptotic, choose an integer d with $\alpha^{d-1} < q < \alpha^d$. The Binet formula shows that, for all sufficiently large j, the sign of $L_{j+s} - qL_j$ is negative if $s < d$ and positive if $s \ge d$. Hence the number of positive index pairs with first index at most N is $\frac{1}{2}N^2 + O_q(N)$. Those with $L_i \le X$ are automatically at most X. Conversely, (8) implies $L_i \le X + q((q+1)(X+q+1)+1) = O_q(X)$ for every positive representation at most X. Thus all contributing first indices satisfy $i \le \log_\varphi X + O_q(1)$. Theorem 12 removes only a finite number of repeated values, giving the result. □

Context. The case $q = 5F_2 = 5$ generalizes the unique representation of positive differences in OEIS A191981. The two coefficient families provide explicit sufficient conditions for uniqueness and for infinite collisions. The classification of all integer coefficients remains open.